

Chp 25, 26, 28, 29

Basic formulas.

Ch25:-

$$q = CV$$

$$q = \epsilon_0 \int \vec{E} \cdot d\vec{A}$$

$$\Delta V = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$C_{eq} = C_1 + C_2 + C_3 \text{ (parallel)}$$

$$1/C_{eq} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \text{ (series)}$$

$$C = \frac{A \epsilon_0}{d} \rightarrow \text{parallel plate capacitor}$$

$$E = \frac{q}{(2\pi r l) \epsilon_0}$$

$$C = \frac{2\pi L \epsilon_0}{\ln(b/a)}$$

Cylindrical capacitors.

$$A = 4\pi r^2$$

$$E = \frac{q}{4\pi r^2 \cdot \epsilon_0}$$

$$V = \frac{q}{4\pi \epsilon_0} \left[\frac{b-a}{ab} \right]$$

$$C = 4\pi \epsilon_0 \cdot \left[\frac{ab}{b-a} \right]$$

Spherical capacitors

$$\frac{1}{2} \epsilon_0 E^2 \Rightarrow \text{Energy density (u)}$$

$$U = \frac{q^2}{2C}$$

$$C = 4\pi \epsilon_0 R \text{ (isolated sphere)}$$

Ch 26:

$$i = \frac{q}{t}$$

$$i = \int \vec{J} \cdot d\vec{A}$$

$$\rightarrow J = \frac{i}{A}$$

proved below

$$\boxed{\vec{J} = (ne) \vec{v}_d}$$

↓

$$q = ne$$

for wire of

$$q = (nAl)e$$

length l .

$$t = \frac{l}{v_d}$$

$$i = \frac{q}{t} = \frac{nA l e}{l/v_d} = \boxed{v_d n A e}$$

$$J = \frac{i}{A} = \frac{(nAe) v_d}{A} = \boxed{(ne) \cdot v_d}$$

$$V = iR$$

$$(\text{resistivity}) \cdot \rho = \frac{E}{J} = \frac{\text{Electric field}}{\text{current density}}$$

$$(\text{conductivity}) \cdot \sigma = \frac{1}{\rho}$$

$$\boxed{R = \frac{V}{I}}$$

$$\rho = \frac{V/L}{i/A} \Rightarrow R = \frac{\rho L}{A}$$

$$\text{from } i = \frac{q}{t} = \frac{nA l e}{t}$$

$$t = \frac{l A n e}{i}$$

Ch 28:

$$\bullet F_B = q (\vec{v} \times \vec{B})$$

$$F_B = F_c$$

$$qvB = \frac{mv^2}{r}$$

$$\bullet r = \frac{mv}{qB}$$

$$s = vt \Rightarrow T = \frac{2\pi m}{qB}, (\omega \text{ \& } f)$$

$$t = \frac{L}{v_d}, q = it = \frac{iL}{v_d}$$

$$F_B = qv_d B = \left(\frac{iL}{v_d} \right) \cdot v_d B \Rightarrow \boxed{iLB = F_B}$$

$$\oint E = B \oint v_d$$

$$E = B v_d$$

$$\frac{V}{l} = \frac{i}{ne} \cdot B$$

$$V = \frac{iB}{ne}$$

$$\boxed{V = \frac{iB}{ne}}$$

$$(\because F_e = F_B)$$

$$v_d = \frac{J}{ne}$$

$$v_d = \frac{i}{ne}$$

$$v_d = \frac{i}{ne}$$

Ch 29:

$$B = \frac{\mu_0 i}{2\pi R} \quad \left(\begin{array}{l} \text{long straight} \\ \text{wire} \end{array} \right)$$

$$\left\{ \mu_0 \Rightarrow 4\pi \times 10^{-7} \text{ Tm/A} \approx 1.26 \times 10^{-6} \text{ Tm/A} \right.$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$$

$$B = \frac{\mu_0 i}{2\pi R} \quad \left(\begin{array}{l} \text{outside the} \\ \text{wire} \end{array} \right)$$

$$B = \frac{\mu_0 i}{2\pi R^2} r \quad \left(\text{inside} \right)$$

Force b/w two wires:

$$\vec{F}_{ba} = i_b (\vec{L} \times \vec{B}_a)$$

$$F_{ba} = \frac{\mu_0 i_a i_b}{2\pi d} L$$

$$B = \mu_0 n i \rightarrow \text{solenoid } (n = N/h)$$

$$B = \frac{\mu_0 i N}{2\pi R} \rightarrow \text{torroid}$$